CHARACTERIZATION OF MICROORGANISMS INVOLVED IN PROCESSES OF ANAEROBIC DIGESTION OF RESIDUAL BIOMASS USING METAGENOMICS TOOLS

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INTRODUCTION

Anaerobic digestion (AD) of residual biomass from the agricultural industry has gained significant interest worldwide due to three fundamental aspects: First, these oxidative processes can reduce both pathogen loads and compounds such as antibiotics, which can alter the native populations of natural systems exposed to waste. Second, AD enables the production of organic fertilizers, which are essential for implementing sustainable agricultural production systems. Third, it focuses on generating biofuels like biomethane and biohydrogen.

The optimal operation of the process requires microbial consortia that participate in various stages of the oxidation-reduction processes. The process is divided into four stages: hydrolysis, acidogenic, acetogenic, and methanogenic. The microorganisms involved may vary depending on the materials being degraded. In the first two phases, bacteria from at least 128 orders across 58 species and 18 genera participate; in the third and fourth phases, 81 species from 23 genera, 10 families, and 4 orders have been identified, primarily consisting of methanogenic bacteria. The goal of this study was to characterize the communities of microorganisms present in each phase by sequencing the 16S rRNA V3-V4 region.

METHODS

We present a preliminary analysis of the bacterial and archaeal composition, including relative abundances, across six bioreactor stages, figure 1. (raw material, hydrolysis, acidogenic, acetogenic, methanogenic, and effluent).

One sample from each of the six stages was sequenced using the Illumina iSeq100, processed, and analyzed using QIIME2. After denoising with DADA2, all samples retained more than 70% of the input reads. Stages 3 and 5 had the lowest and highest number of Amplicon Sequence Variants (ASVs), respectively; however, Stages 2 and 5 exhibited the lowest and highest Shannon index values. Sequencing depth exceeded 126,000 reads for all samples, with a minimum depth of 100,000 reads.

Figure1. Sampling areas. 1. Inoculum sludge. 2. Feed. 3. First sampling area. 4. Second sampling area. 5. Third sampling area. 6. Effluent (not shown).

RESULTS

Changes in bacterial and archaeal composition were observed throughout the bioreactor stages (figure 2). For instance, during the initial stages (1 and 2), Firmicutes and Halobacterota were the most abundant phyla, representing 22.2% and 43.3%, respectively. By stage 3, Firmicutes maintained a high abundance of 42.8%, while Proteobacteria emerged as the second most abundant phylum at 42.7%. In stage 4, Proteobacteria dominated at 49.2%, while Firmicutes decreased to 11.1%, and Chloroflexi increased to 13.2%, becoming the second most abundant phylum. In the final stages of the bioreactor (5 and 6), notable changes were observed in Thermoplasmatota, Actinobacteriota, and Synergistota (figure 3).

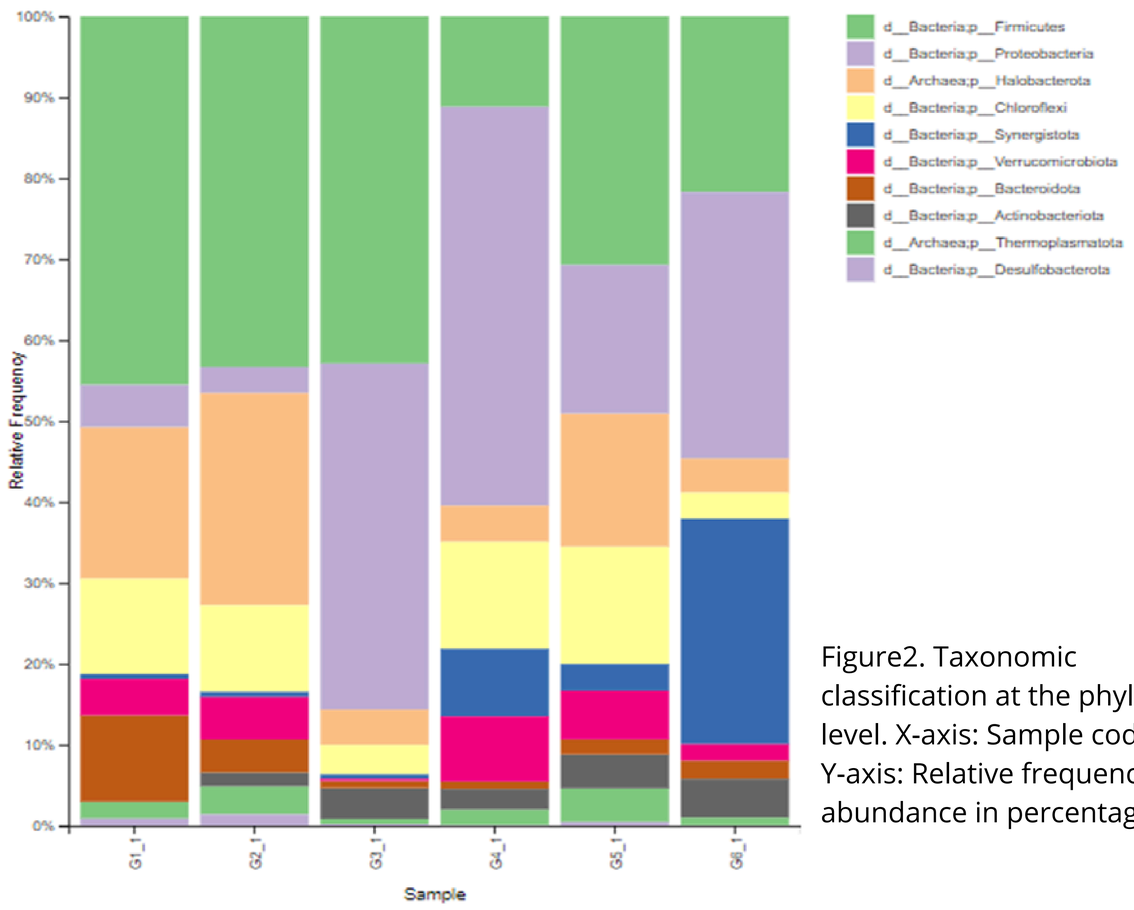

Figure2. Taxonomic classification at the phylum level. X-axis: Sample code. Y-axis: Relative frequency of abundance in percentage.

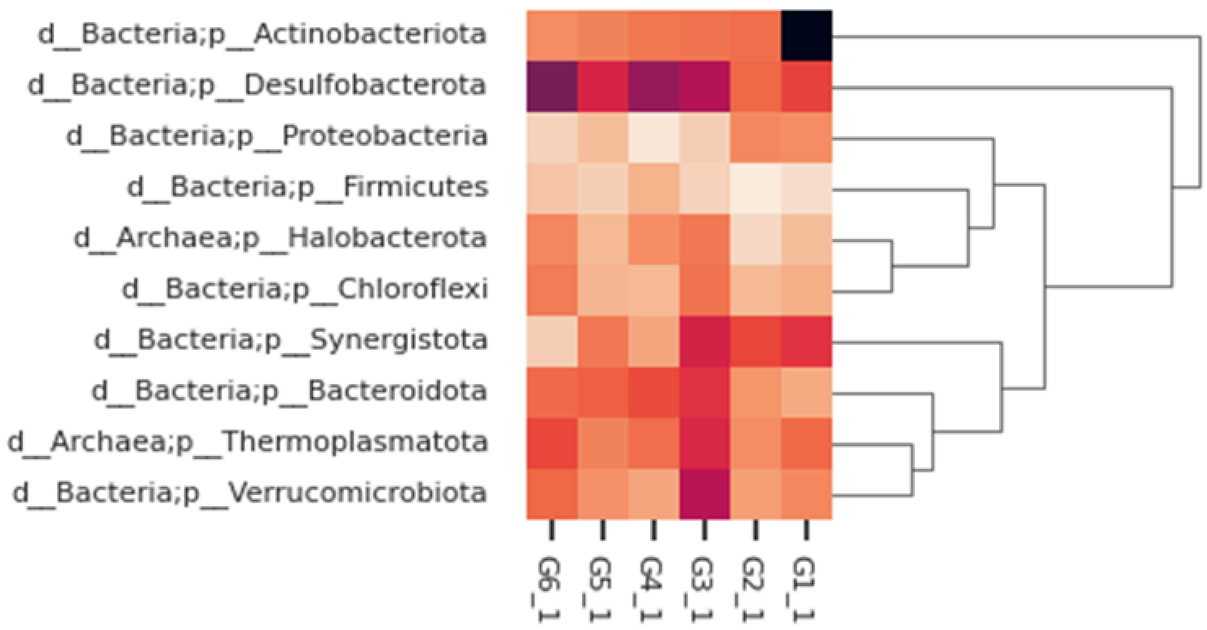

Figure3. Clustering analysis at the phylum level Dark: low frequency. Light: high frequency.

CONCLUSIÓN

This knowledge is important to optimize the efficiency of bioenergy production, improve the quality of the products obtained and ensure that the process remains stable and meets environmental standards. Furthermore, it can allow the implementation of more effective management strategies and the development of new technologies for waste treatment and renewable energy production.